

Eigenvalues and inverse problems

Find a near-largest nonnormal eigenvalue using few matrix-vector products

Difficulty: challenging **Importance:** interesting to the community

Status: **Lean verified**

Topic: nonsymmetric eigenvalue computation; query complexity

Rating rationale: Challenging reflects a query bound for general nonnormal matrices, whose eigenvalues need not be controlled by power iteration; community impact is the complexity of large nonsymmetric eigenproblems.

Resolution — 2026-09-11

Affirmative resolution by Matthew J. Colbrook (Department of Applied Mathematics and Theoretical Physics, University of Cambridge). See the [complete manuscript](#), **Theorem KE-03, sections 1–5** ([PDF · LaTeX](#)), prepared 11 September 2026.

The algorithm uses $O(\varepsilon^{-2}[1 + \log(nK)])$ exact matrix-vector queries, with success probability at least 0.997, for every input in the displayed model. It supplies both eigenvalue-location guarantees using the given condition bound K and finite exact arithmetic between queries. The result bounds query count, not total runtime, bit complexity or floating-point error.

The original proof draft was generated in a ChatGPT conversation. A separate Codex agent independently verified the full proof and its match to the exact target on 11 September 2026: [detailed PASS review](#). The review records a hash of the unchanged proof text. This is independent agent verification, not external human peer review or formal certification. [The submission history and diagnostic record](#) preserve the initial solution claim. The ratings above are historical, and the earlier literature checks below are retained.

Lean proof and verification evidence

The [complete Lean proof](#) formalizes the entire original exact-query target. Its declaration `NLA.KE03.complete_query_algorithm` proves that the concrete algorithm terminates on every seed and uses at most $32768[1 + \log(nK)]/\varepsilon^2$ queries. With probability at least 99/100, one actual eigenvalue satisfies both required inequalities. All positive dimensions, supplied condition bounds, allowed tolerances, spectral scales, repeated eigenvalues and ties are included.

The formalization uses an integer random grid and exact finite counting, positive-rational enumeration for radius approximation, and a rational circle mesh. [The statement correspondence](#) and [proof guide](#) explain the complete algorithm, Euclidean norms, uniform seed law, and query accounting. No diagonalizer or spectral information is an algorithm input. Total runtime, bit complexity and floating-point stability remain outside the original target.

The project pins Lean 4.33.1 and Mathlib `0df444a360eaa60ab8c11dca51a86af692955474`. The [verification record](#) includes reproduction commands, source hashes, local build and transitive axiom logs, and the successful fresh Ubuntu Comparator run of 28 September 2026. The complete theorem uses only `propext`, `Classical.choice` and `Quot.sound`; the independent Challenge placeholder is excluded from its dependencies.

The mathematical resolution remains credited to Matthew J. Colbrook. OpenAI Codex agents authored the Lean implementation. Two separate, nonauthor agents approved the full statement and proof; [their reports](#) disclose their scope and evidence. This is AI-agent review, with no claim of external human peer review. The complete proof passed statement comparison, permitted-axiom checks, default-kernel replay and rejection controls; the retained evidence supports **Lean verified** status.

Context and notation

All norms are Euclidean vector norms or their induced matrix norms.

Problem statement

An unknown diagonalizable $A \in \mathbb{C}^{n \times n}$ is available only through exact queries $v \mapsto Av$. A number $K \geq 1$ is supplied, with the promise that $A = V\Lambda V^{-1}$ for some diagonal Λ and $\|V\|_2\|V^{-1}\|_2 \leq K$. Assume $\rho(A) = \max_{\mu \in \sigma(A)} |\mu| > 0$; neither the eigenvalues nor V are supplied.

Do universal constants $C, a, b > 0$ and a randomized algorithm exist that, for every such input and every $\varepsilon \in (0, 1/2)$, returns a complex number z with probability at least 0.99 such that

$$\text{there exists } \mu \in \sigma(A) : \quad |\mu| \geq (1 - \varepsilon)\rho(A), \quad |z - \mu| \leq \varepsilon\rho(A),$$

using at most $C[1 + \log(nK)]^a \varepsilon^{-b}$ matrix-vector queries? The algorithm may perform finite exact arithmetic between queries; the requested bound concerns query count. Products with A^* and solutions of shifted linear systems are not additional oracle operations. Supplying K and fixing a success probability make the computational formulation explicit.

References

Amsel et al., [workshop report](#), Problem 3.9. Shah, Srivastava, and Zeng, *Sparse Pseudospectral Shattering*, [first version](#), §1.3, Theorem 1.5 and its power-iteration argument, supplies the related spectral-radius estimate cited by the workshop.

Earlier status check — 2026-09-08

Searches for *nonnormal eigenvalue query complexity 2026* and *nonnormal spectral radius matrix-vector algorithm 2026* found no resolution. The radius estimate controls a nonnegative scalar and allows backward error; it does not locate a near-extremal eigenvalue in the complex plane as required here. The [April 2026 third version](#) replaces that application with a GMRES application (§6), so the version-specific locator above is intentional. The workshop still poses Problem 3.9 in its August update, separately from its resolved general Ritz-compression question.

Audit update — 2026-09-10

Rechecked [workshop version 3](#), Problem 3.9. Searches for nonnormal largest-eigenvalue matrix-vector query bounds found no theorem or lower bound settling the displayed question. Hermitian or singular-value oracle results do not answer it.